INCH-POUND

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March 8, 1991
SUPERSEDING
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March 27, 1987

COMMERCIAL ITEM DESCRIPTION

RACK, STORAGE, LOADED PALLET, GALVANIZED

The General Services Administration has authorized the use of this commercial item description.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers loaded pallet, galvanized storage racks in the following sizes (see 5.2):

Size 1 - 30-inch wide upright frame
Size 2 - 42-inch wide upright frame

1.2 CID based part identification number (PIN). A document based PIN to identify sizes of storage racks is included in 5.4.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. Racks shall consist of the following galvanized steel constructed components: removable wire mesh decking, upright frame posts, shelf beams with safety locks, cross braces for front-to-rear post support, back-to-back connectors, leveling shims, and floor bearing plates. The pallet rack components may be manufactured from hot-rolled steel, or cold-formed steel. The design of all hot-rolled steel components shall conform to the American Institute of Steel Construction (AISC), Manual of Steel Construction Allowable Stress Design. Cold-formed steel components shall conform to the American Iron and Steel Institute (AISI), Specification for the Design of Cold-Formed Structural Members. The load bearing components shall be designed for easy side-to-side rack assembly during rack erection without the use of tools other than a soft hammer. These components shall be constructed to produce a rigid and stable rack capable of withstanding a minimum shelf load of 6,000 pounds.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Commander, U.S. Amry Natick Research, Development, and Engineering Center, ATTN: STRNC-ES, Natick, MA 01760-5017.

AMSC N/A FSC 7125

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The upright frame unit shall have a minimum load capacity of 24,000 pounds. An assembled rack unit shall consist of two upright frames attached to a minimum of two shelf units. The storage rack shall measure approximately 108 inches long by 108 inches high.

2.2 Upright frame. The upright frame shall be welded construction and shall consist of two vertical posts, minimum 3-inch by 3-inch angle dimensions. The two vertical posts shall be braced by a minimum of three bracing bars consisting of "X" type bars between two horizontal bars for each brace and secured to the floor bearing plates. Posts shall be interchangeable in any assembled position. Each post shall have openings to accommodate shelf-beam connectors at corresponding heights. The upright frame for the size 1 rack shall be 30 inches wide by 108 inches high. The upright frame for the size 2 rack shall be 42 inches wide by 108 inches high.

2.3 Shelf unit. The shelf unit shall have a minimum safe load capacity of 6,000 pounds, and shall consist of two 108-inch long load beams with two sections of wire mesh decking of equal length. The ends of the beams shall have either a multiple prong slip fit or slip joint to easily fasten to post connectors. Each beam-to-post connector shall have a safety lock which will lock the beam to the posts. The beams shall be approximately 4-1/2 inches high by 3 inches deep, or the depth of the beam shall be compatible with the width of the upright frame (see 5.2).

2.3.1 Wire mesh decking. Two sections of wire mesh decking of equal length shall be provided for each shelf unit. The wire mesh decking shall be constructed of 2-gauge minimum steel wire, and shall have a maximum 2-inch by 2-inch diamond or square shaped meshwork. For pallet racks with box type load beams, four minimum 14-gauge flared channel supports a minimum of 2 inches wide by 1-3/8 inches deep shall be welded to each section of wire mesh decking. For pallet racks with step-type load beams, four minimum 12-gauge "U" shaped supports, a minimum of 1 inch wide by 1-3/8 inches deep shall be welded to each section of wire mesh decking. The front and rear edges of the decking shall be waterfall design (flanged) to fit over load beams. The wire mesh decking shall be removable from the rack shelf (see 5.2).

2.4 Back-to-back connectors. Back-to-back connectors shall be furnished to securely attach the backs of the units to form a double row of racks. The connectors provided shall be either the slip-on boltless type or the bolt, washer, and nut variety. The connectors shall measure 12 inches long and shall have strength to withstand all forces applied to the connected rack units during normal loading and unloading operations (see 5.2).

2.5 Leveling shims. A sufficient amount of leveling shims shall be provided with each rack unit. The shim shall be a minimum of 12-gauge minimum steel and of the same size as the floor bearing plate. The shim anchor or lock with the floor bearing plate in such a manner as not to separate during normal storage operations.

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2.6 Standards compliance. The rack components shall comply with ASTM A 123 for hot galvanized dipped coating.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.3 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Packaging. The storage rack shall be packaged, packed, and marked in accordance with the domestic or overseas requirements of ASTM D 3951, as applicable.

5. NOTES

5.1 Intended use. The boltless, steel, pallet racks are intended for storing palletized loads.

5.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size rack required (see 1.1).
c. Number of upright frames (see 2.2).
d. Number of load beams (see 2.3).
e. Number of units of wire mesh decking (see 2.3.1).
f. Number of back-to-back connectors (see 2.4).

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5.3 Sources of documents.

5.3.1 Sources for nongovernment association documents.

ASTM A 123 - Zinc (Hot-Galvanized) Coatings on Products Fabricated From Rolled, Pressed, and Forged Steel Shapes, Plates, Bars and Strip

ASTM D 3951 - Standard Practice for Commercial Packaging, are available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

AISC - Manual for Steel Construction, Allowable Stress Design, is available from:

The American Institute of Steel Construction (AISC)
1 East Wacker Drive
Suite 3100
Chicago, IL 60601-2001

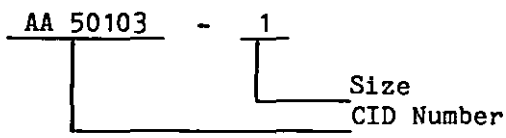
AISI - Specification for the Design of Cold-Formed Steel Structural Members, is available from:

The American Iron and Steel Institute (AISI)
1133 15th N.W. Street
Suite 300
Washington, DC 20005-2701

5.3.2 Sources of governmental documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.4 Part identification number (PIN). The PIN's to be used for the storage racks required by this CID are created as follows:



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MILITARY INTERESTS:

Custodian

Army - GL

Review Activity

Army - TS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project 7125-0131

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